Mohammad Saad Nathani

Vapi, Gujarat, India saadnathani2005@gmail.com | LinkedIn | GitHub | Portfolio

Applied ML/DL engineering student with hands-on experience across deep learning, RAG systems, and agentic AI tooling. ACM Student Chapter Chair, NMIMS Shirpur.

Education

SVKM NMIMS Mukesh Patel School of Technology Management and Engineering 2023 – 2027
Bachelor of Technology in Artificial Intelligence and Machine Learning CGPA: **3.62**/4.00 (6th Sem)
St. Xavier's High School, Vapi 10+2

Experience

CN Automation LLP [Link] Jun 2026 – Jul 2026
Engineering Intern (On-site)

- Built a local RAG assistant for industrial manuals using Docling, Ollama (qwen2.5:7b, nomic-embed-text) and ChromaDB, deployed via Docker with a Streamlit UI.
- Boosted answer accuracy with hierarchical chunking and hybrid BM25 + vector retrieval, returning page-level citations and diagrams.

Projects

- ESG Survey Dashboard – Mirae Asset Foundation [Link] Mar 2026
Python, Power BI
Built the data pipeline and custom ESG scoring logic behind a survey analytics dashboard, processing 1,641 responses across 21 villages in Shirpur district; the scoring system classified the district in the "Safe" ESG category, indicating acceptable compliance.

- ShopSense AI – AI-Powered Shopping Assistant Jul 2025
Python, OpenAI API, Llama 7B (local), Chrome Extension APIs
Built a browser extension that scrapes and structures product data from e-commerce pages, using an LLM to answer natural-language shopping questions on compatibility, use-case fit, and recommendations across categories like tech and skincare. Supported both OpenAI's API and a locally-hosted Llama 7B model, giving users a free/offline option without requiring an API key.

- FloatChat – Natural Language Climate Data Query System [Link] Aug 2025
Python, FAISS, PostgreSQL, NetCDF/Xarray, Claude, LangChain, MCP
Architected and led an end-to-end pipeline for ingesting daily Argo NetCDF ocean datasets into PostgreSQL; developed an MCP server exposing the data to an LLM agent for natural-language queries, rendering results as interactive maps and replacing manual report generation.

- Holistically-Nested Edge Detection [Link] Oct 2025
TensorFlow, VGG16, OpenCV
Reimplemented HED from Caffe to TensorFlow/Keras with a VGG16 backbone, five deeply supervised side outputs, and a learned fusion layer trained with class-balanced loss on BSDS500. Published as `hed-tensor` on PyPI with bundled pretrained weights and a three-line inference API.

Capstone Project

Surveillance Anomaly Detection with VL-JEPA *Ongoing*
PyTorch, Hugging Face
Developing a surveillance anomaly detection system using the VL-JEPA architecture, training on annotated UCF-Crime data to detect anomalies and generate natural-language alerts for authorities designed with edge-device constraints in mind. (team capstone, 15-week timeline).

Achievements

- **1st Place** — AISEhack Theme 2: Pollution Prediction (ANRF, IBM, IITD, IIITH) [Link] Apr 2026
- **2nd Place** — Yugantar 2025 (Smart India Hackathon Internal Round) [Link] Sept 2025
- **Remote Finalist** — HackRx 6.0 (Bajaj Finserv Hackathon 2025) Aug 2025

Skills

Languages: Python, C++
AI/ML Core: PyTorch, LangChain, LangGraph RAG,MCP, Claude SDK, OpenAI
Libraries: NumPy, Pandas, Seaborn, Matplotlib, Plotly, OpenCV, NLTK, Scikit-learn, XGBoost
Frameworks/Tools: Streamlit, FastAPI
Data/BI: MySQL, Oracle SQL, Tableau, Power BI, SSIS, SAS

Certifications

Machine Learning - Banaras Hindu University (**Score: 90%**) [Link] May 2025
Model Context Protocol: Advanced Topics - Anthropic [Link] July 2026
Excel: Beginner to Advanced - Udemy [Link] Dec 2024

Leadership & Extracurricular

Chair, ACM Student Chapter – NMIMS MPSTME Shirpur [Link] Oct 2025
Led chapter operations and organized technical events, workshops, and student engagement initiatives.